

Extrapolating from Mines Paris-PSL survey values to VALESOR's tool

Daniel Herrera-Araujo*, Maria Kostopoulou

This version: December 2025

1 From Country Data to Final 2020 PPP Values

This note describes the steps used to move from the raw country data to the final estimates of VSL, and VOLY in 2020 PPP euros for each country/region in the dataset.

1.1 Data Inputs

We use three main building blocks:

1. **Country list (INPUT1).** A list of countries and regional aggregates (e.g. Pan-European, EU27, Western Balkan, EECCA, UNECE Europe, UNECE Caucasus + Asia, EFTA, and individual countries such as France, Italy, Spain, etc.). For each, we ultimately retain:
 - GDP per capita, PPP;
 - Population;
 - Two monetary values: VSL, VOLY (for morbidity).
2. **Economic data (GDP per capita, PPP).** For all countries/regions in INPUT1, we match GDP per capita (PPP, in 2025 prices) from the PPP table (World Bank-style data). This yields:

$$y_i = \text{GDP per capita, PPP (2025 prices) for country } i.$$

*Herrera-Araujo: Mines Paris-PSL, (Corresponding author: daniel.herrera@minesparis.psl.eu);
Kostopoulou: Mines Paris-PSL

3. **Population data.** For the same countries/regions, we match population figures:

$$P_i = \text{population of country/region } i.$$

Throughout, we denote by y_i the GDP per capita (PPP, 2025 prices) of country i and by P_i its population.

1.2 Six-Country Values for VSL and VOLY

The monetary values for the six EU countries (France, Italy, Spain, Sweden, Poland, Greece) come from a representative willingness-to-pay (WTP) survey fielded separately in each country. The survey elicits WTP for reducing both *mortality risk* (the probability of becoming ill and not surviving) and *morbidity risk* (the probability of becoming ill and surviving). It relies on a discrete choice experiment (DCE) design in which respondents repeatedly compare a baseline option, featuring higher risks and no additional cost, to two safer alternatives (A and B) that reduce risks but involve higher costs. Each respondent faces ten choice sets, and we have about 650 respondents per country at this stage.¹

From the estimated DCE models, we recover country-specific WTP schedules for marginal changes in fatal and non-fatal risks. Combining these schedules with the underlying risk changes yields four monetary measures for each country i :

- the value per statistical life, VSL_i ;
- the value per life-year (VOLY), $VOLY_i$.

All values are expressed in millions of 2025 PPP euros. We match these survey-based values to the same population and GDP per capita data (P_i, y_i) used in the rest of the tool.

Table 1 reports the resulting VSL, morbidity-related VOLY for each of the six countries, together with their populations. The last row shows the pooled value obtained by aggregating countries with population weights; this row corresponds exactly to the population-weighted averages formalized in the next subsection.

¹The initial realised sample comprised 1,435 respondents per country. After applying standard data-quality checks (e.g. excluding speedsters, respondents failing comprehension or attention checks, and protest responses), the final sample is reduced to around 650 respondents per country.

Country	Population	VSL	VOLY
France	66,186,944	5.950	0.1520
Italy	59,947,538	3.931	0.1110
Spain	47,715,662	7.200	0.1800
Sweden	10,353,687	7.335	0.1730
Poland	38,171,012	3.867	0.1050
Greece	10,699,369	3.200	0.0807
Pop-weighted		5.280	0.130

Table 1: VSL and morbidity-related VOLY for six countries and their pooled population-weighted average. All values are expressed in millions of 2025 PPP euros.

1.3 Population-Weighted Averages (Anchors)

We use the six-country values in Table 1 to construct population-weighted averages that serve as anchors for the scaling exercise. Let $S = \{\text{France, Italy, Spain, Sweden, Poland, Greece}\}$ denote the set of six countries.

Total population of the six countries.

$$P^{(6)} = \sum_{i \in S} P_i.$$

Population-weighted averages. For each metric $VALUE \in \{VSL, VOLY\}$ we compute:

$$VALUE^{PW} = \frac{\sum_{i \in S} P_i \cdot VALUE_i}{\sum_{i \in S} P_i} = \frac{\sum_{i \in S} P_i \cdot VALUE_i}{P^{(6)}}.$$

In particular:

$$VSL^{PW} = \frac{\sum_{i \in S} P_i \cdot VSL_i}{P^{(6)}},$$

$$VOLY^{PW} = \frac{\sum_{i \in S} P_i \cdot VOLY_i}{P^{(6)}}.$$

Numerically, the pooled row in Table 1 corresponds to:

$$VSL^{PW} \approx 5.28, \quad VOLY^{PW} \approx 0.13,$$

all in millions of 2025 PPP euros. We treat these as the “representative” six-country anchors that we subsequently scale to other countries.

1.4 Population-Weighted GDP per Capita for the Six Countries

To scale values for all countries based on income differences, we construct a population-weighted GDP per capita for the same six countries:

$$y^{PW} = \frac{\sum_{i \in S} P_i \cdot y_i}{\sum_{i \in S} P_i} = \frac{\sum_{i \in S} P_i \cdot y_i}{P^{(6)}}.$$

Using the six countries' GDP per capita (PPP, 2025 prices) and populations, we obtain approximately

$$y^{PW} \approx 43,516 \text{ euros (PPP per capita, 2025 prices).}$$

This y^{PW} provides the income reference point for the scaling formulas in the next subsection.

1.5 Scaling Values to All Countries Using GDP per Capita

We then project the four anchors to all countries/regions in INPUT1 using the ratio of each country's GDP per capita to the six-country population-weighted GDP per capita.

For each country i with observed GDP per capita y_i , and for each metric $\text{VALUE} \in \{\text{VSL}, \text{VOLY}\}$, we assume:

$$\text{VALUE}_i = \text{VALUE}^{PW} \times \left(\frac{y_i}{y^{PW}} \right)^\varepsilon,$$

where:

- VALUE^{PW} is the population-weighted anchor from the six countries,
- y^{PW} is the six-country population-weighted GDP per capita,
- ε is the income elasticity of willingness-to-pay.

In our baseline, we set:

$$\varepsilon = 1,$$

so the formula simplifies to:

$$\text{VALUE}_i = \text{VALUE}^{PW} \times \frac{y_i}{y^{PW}}.$$

We apply this separately for:

$$\text{VSL}_i, \quad \text{VOLY}_i,$$

which yields four new columns in the country-level dataset (still in millions of 2025 PPP euros), e.g. `VSL_scaledPPP`, `VOLY_scaledPPP`.

For countries/aggregates where GDP per capita is missing (e.g. Pan-European, Western Balkan, EECCA, UNECE Europe, UNECE Caucasus + Asia, EFTA), these scaled values are left as missing at this stage.

1.6 Aggregating Countries into Areas and Area-Level GDP per Capita

In addition to country-level values, the VALESOR tool reports information for regional aggregates: EU27, Western Balkan, EECCA, UNECE Europe, UNECE Caucasus + Asia and EFTA.²

For each area a , with member countries $i \in a$, we compute the *population-weighted area GDP per capita* as:

$$y_a^{PW} = \frac{\sum_{i \in a} P_i \cdot y_i}{\sum_{i \in a} P_i} = \frac{\sum_{i \in a} P_i \cdot y_i}{P_a},$$

where

$$P_a = \sum_{i \in a} P_i,$$

is the total population of area a . Both y_i and y_a^{PW} are expressed in PPP euros at 2025 prices.

These regional GDP per capita figures are computed purely from the country-level GDP per capita and population data; they do not change the country-level scaling described above, but are used whenever the tool needs to report or visualise results at the level of these regions rather than individual countries.

²Membership is taken from the file *List of countries and country groups* used in VALESOR. The areas are defined as follows (country names as in the membership file). EU27: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Poland, Portugal, Romania, Slovak Republic, Slovenia, Spain, Sweden. Western Balkan: Albania, Bosnia and Herzegovina, North Macedonia, Montenegro, Serbia. EECCA: Armenia, Azerbaijan, Belarus, Georgia, Kazakhstan, Kyrgyzstan, Republic of Moldova, Russia European part, Tajikistan, Turkmenistan, Ukraine, Uzbekistan. UNECE Europe: all EU27 countries plus Albania, Belarus, Bosnia and Herzegovina, Iceland, North Macedonia, Republic of Moldova, Montenegro, Norway, Russia European part, Serbia, Switzerland, Ukraine, United Kingdom. UNECE Caucasus + Asia: Armenia, Azerbaijan, Georgia, Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, Turkey, Uzbekistan. EFTA: Iceland, Norway, Switzerland. Countries listed in the column *Merged with small countries and dependencies* (e.g. Austria with Liechtenstein; France with Andorra, Monaco, Guernsey, Jersey; Italy with San Marino and Vatican City; Spain with Gibraltar; United Kingdom with Isle of Man) are treated as a single unit; we use the GDP per capita and population of the main country for the merged group because separate PPP and population data for the micro-states are not available in our sources.

1.7 Converting from 2025 PPP to 2020 PPP Euros

All WTP values reported above (VSL, VOLY) are expressed in millions of PPP euros at the price level that we label “2025 PPP”. To express these values in 2020 PPP euros in a transparent and reproducible way, we proceeded in three steps.

Step 1: Choice of price index and series. We approximate the change in the general price level between 2020 and 2025 using the euro area Harmonised Index of Consumer Prices (HICP), overall index, with base year 2015=100. Concretely, we use the annual HICP index for the euro area (changing composition) from Eurostat’s dataset `prc_hicp_aind` (all-items HICP, index 2015=100).³

At the time of coding, the latest annual index available for the euro area was for 2024, not 2025. We therefore approximate the 2025 price level with the 2024 annual HICP index.

Step 2: Numerical values and deflator. In the code, we set:

$$P_{2020} = 99.81, \quad P_{2024} = 126.08,$$

where P_{2020} and P_{2024} denote the annual average euro area HICP indices (2015=100) for 2020 and 2024, respectively, taken from the Eurostat series described above. Using these two numbers, we define a common deflator to move from 2025/2024 PPP euros to 2020 PPP euros:

$$d \equiv \frac{P_{2020}}{P_{2024}} = \frac{99.81}{126.08} \approx 0.79164.$$

Operationally, we treat our “2025 PPP” values as being at the 2024 HICP price level, and then multiply by d to re-express them in 2020 PPP euros.

Step 3: Applying the deflator to the WTP metrics. For any WTP value X_i^{2025} (in millions of PPP euros) for country i , measured at the 2025/2024 price level, we construct the corresponding value in 2020 PPP euros as:

$$X_i^{2020} = X_i^{2025} \times d.$$

³Eurostat, “HICP – annual data (average index and rate of change) (`prc_hicp_aind`)”, overall index, euro area (changing composition), accessed via the Eurostat database.

We apply this transformation uniformly to the four WTP metrics for every country:

$$\begin{aligned} VSL_i^{2020} &= VSL_i^{2025} \times d, \\ VOLY_i^{2020} &= VOLY_i^{2025} \times d. \end{aligned}$$

In the dataset, these appear as four additional columns. We do not adjust GDP per capita or population; only the WTP measures are deflated from 2025/2024 PPP to 2020 PPP euros.

1.8 Alternative VSL based on OECD (2025) Meta-Analysis

In addition to the VALESOR-specific VSL column anchored on the six-country survey, we construct a second VSL column based on the recent OECD VSL meta-analysis for EU countries. According to the OECD (2025) meta-analysis,⁴ the population-weighted mean VSL for EU27 countries is 8.4 million 2020 USD, with a 95% confidence interval of 7.0–9.8 million 2020 USD. Using the PPP-corrected exchange rate between USD and euro (EU27) in 2020, equal to 0.766 (euro per USD), this corresponds to:

$$VSL_{2020}^{\text{OECD,EU27}} = 8.4 \times 0.766 \approx 6.4,$$

million 2020 euros (PPP) per statistical life. We use this value as an alternative EU27-wide anchor for VSL.

Let y_i^{2020} denote GDP per capita (PPP) in 2020 for country i , and let y_{EU27}^{2020} denote GDP per capita (PPP) in 2020 for the EU27 as a whole. In practice, we obtain y_i^{2020} and y_{EU27}^{2020} by applying the same HICP deflator d described above to the 2025 PPP GDP per capita values:

$$y_i^{2020} = y_i^{2025} \times d, \quad y_{EU27}^{2020} = y_{EU27}^{2025} \times d.$$

Assuming an income elasticity of willingness-to-pay $\varepsilon = 1$, we scale the OECD anchor to each country i using the ratio of its GDP per capita to the EU27 GDP per capita:

$$VSL_{i,2020}^{\text{OECD}} = VSL_{2020}^{\text{OECD,EU27}} \times \left(\frac{y_i^{2020}}{y_{EU27}^{2020}} \right)^\varepsilon = 6.4 \times \frac{y_i^{2020}}{y_{EU27}^{2020}}.$$

This yields a second VSL series, expressed directly in millions of 2020 PPP euros, that is fully comparable across countries and consistent with the OECD meta-analytic evidence for

⁴OECD (2025), *Evaluating the Economic Benefits of Environmental Policies that Save Lives*. See, in particular, Table 6.1 and the associated documentation, available at: <https://www.oecd.org/en/events/2025/10/evaluating-the-economic-benefits-of-environmental-policies-that-save-lives.html>.

EU27. In the dataset, we store this as an additional column, for example:

$$\text{VSL_OECD_2020PPP}_i = \text{VSL}_{i,2020}^{\text{OECD}}.$$

This OECD-based VSL column is not used to scale VOLY values; it is included to facilitate comparisons and sensitivity analyses relative to a benchmark grounded in the broader OECD literature.

1.9 Final Dataset Structure

The final country-level dataset contains, for each country/region in INPUT1:

1. Country identifier: `Country`;
2. GDP per capita, PPP (2025 prices): `GDP_per_capita_PPP`.
3. Population: `Population`.
4. Scaled 2025 PPP values (in millions): `VSL_scaledPPP`, `VOLY_scaledPPP`.
5. Scaled 2020 PPP values (in millions): `VSL_scaled_2020PPP`, `VOLY_scaled_2020PPP`.
6. OECD-based VSL benchmark (in millions, 2020 PPP): `VSL_OECD_2020PPP`.

This provides a transparent chain from the original six-country values and country GDP and population data to harmonised 2020 PPP estimates of VSL, and morbidity-related VOLY for all countries and regions in the analysis, together with an additional VSL benchmark grounded in the OECD (2025) meta-analysis.